# Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

**AY: 2026-27**

| | | | |
|---|---|---|---|
| **Class:** | **TE-AI&DS** | **Semester:** | **V** |
| **Course Code:** | **2015119** | **Course Name:** | **Agile SoftwareDevelopment& DevOps Lab** |

| | |
|---|---|
| **Name of Student:** | **Chirag Raut** |
| **Roll No. :** | **85** |
| **Experiment No.:** | **01** |
| **Title of the Experiment:** | **To study and Implement Agile Software Project lifecycle using Jira with DevOps Pipeline Integration.** |
| **Date of Performance:** | |
| **Date of Submission:** | |

## Evaluation

| Performance Indicator | Max. Marks | Marks Obtained |
|---|---|---|
| Performance | 5 | |
| Understanding | 5 | |
| Journal work and timely submission | 10 | |
| Total | 20 | |

| Performance Indicator | Exceed Expectations (EE) | Meet Expectations (ME) | Below Expectations (BE) |
|---|---|---|---|
| Performance | 4-5 | 2-3 | 1 |
| Understanding | 4-5 | 2-3 | 1 |
| Journal work and timely submission | 8-10 | 5-8 | 1-4 |

**Checked by**

**Name of Faculty : Ms. Kshitija Gharat**

**Signature :**

**Date :**

**Aim :** To study and implement the Agile Software Project Lifecycle using Jira by managing Scrum tasks, tracking development progress, integrating GitHub for version control, and implementing a DevOps CI pipeline using GitHub Actions.

**Objectives:**

1. To understand the Agile Software Development lifecycle and Scrum methodology.
2. To create and manage a software project using Jira.
3. To create and track Scrum work items such as user stories and tasks using Jira.
4. To use Git and GitHub for source code management and version control.
5. To create feature branches for implementing individual requirements.
6. To create Pull Requests and merge completed changes into the main branch.
7. To integrate GitHub with Jira for tracking development activities.
8. To configure a Continuous Integration (CI) pipeline using GitHub Actions.
9. To automate testing and verification of the application whenever code changes are pushed.
10. To understand the complete workflow from Jira task creation to code development, CI execution, Pull Request, merging, and task completion.

**Theory :**

**1. Agile Software Development**

Agile is an iterative and incremental approach to software development in which the project is divided into small, manageable units of work. Development is performed in short iterations, allowing continuous feedback, testing, and improvement.

Agile emphasizes:

- Customer and stakeholder collaboration
- Continuous delivery of working software
- Adaptability to changing requirements
- Frequent feedback
- Team collaboration
- Continuous improvement

**2. Scrum**

Scrum is an Agile framework used to manage software development projects. The project work is divided into smaller tasks called work items or user stories and organized into Sprints.

A typical Scrum workflow is:

**Product Backlog → Sprint Planning → Development → Testing → Review → Completion**

In this experiment, Jira is used to create and manage Scrum work items and track their status through different stages such as **To Do, In Progress, and Done**.

**3. Jira**

Jira is a project management and issue-tracking tool widely used for Agile software development. It allows teams to create user stories, tasks, bugs, and other work items and monitor their progress.

Jira provides features such as:

- Product backlog
- Scrum board
- Sprint management ● Work-item tracking
- Development integration
- Progress monitoring

In this experiment, a **Student Management System** project is managed using Jira.

**4. Git and GitHub**

Git is a distributed version control system used to track changes in source code. GitHub provides a cloud-based platform for storing Git repositories and collaborating on software projects.

A feature branch can be created for each task so that development can be performed independently without directly modifying the main branch.

Typical workflow:

**Create Branch → Modify Code → Commit → Push → Pull Request → Review → Merge 5.**

**Pull Request**

A Pull Request (PR) is used to propose changes from one branch to another. It allows developers to review and discuss code changes before they are merged into the main project branch.

In this experiment, Pull Requests were created for Scrum tasks such as student validation and student email validation and were subsequently merged into the main branch.

**6. DevOps and Continuous Integration**

DevOps combines development and operations practices to automate and improve the software delivery process.

Continuous Integration (CI) is a DevOps practice in which developers frequently integrate their code into a shared repository. Automated workflows can then build and test the application to detect errors early.

In this experiment, **GitHub Actions** is used to implement the CI pipeline. Workflow files are stored inside:

.github/workflows/

The pipeline can automatically execute tasks such as:

- Setting up the Python environment
- Installing dependencies
- Running automated tests
- Verifying the application after code changes

**7. Jira–GitHub Integration**

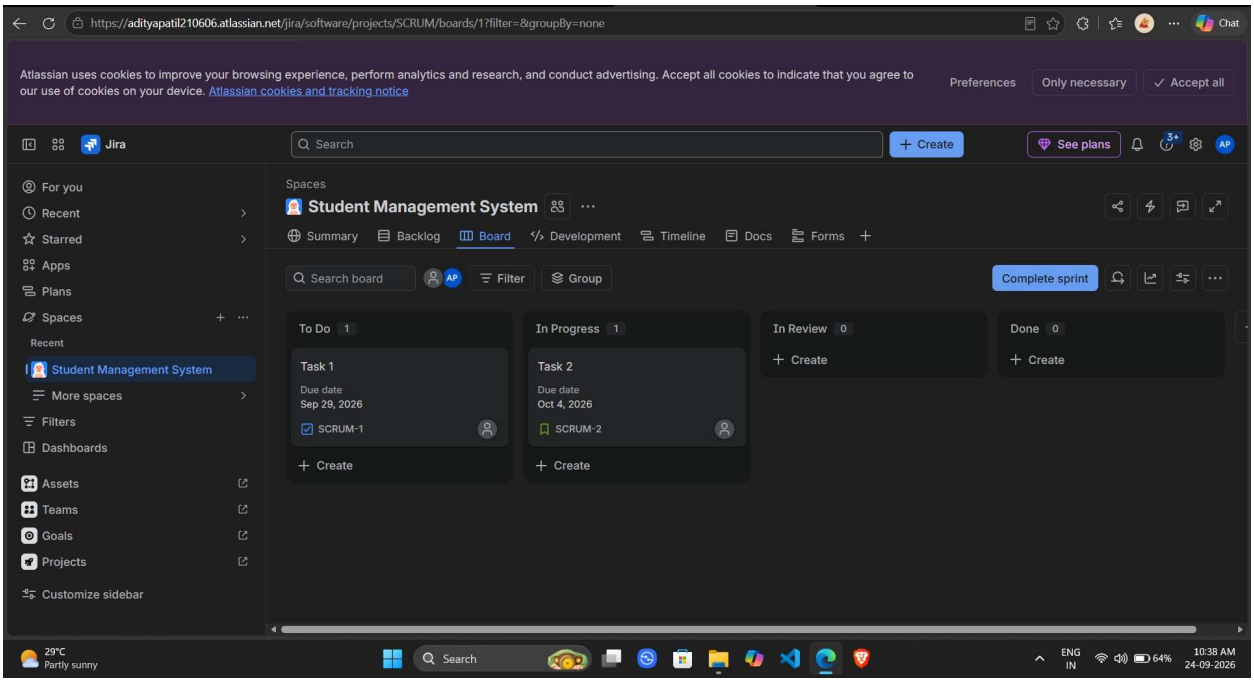
Jira can be integrated with GitHub so that development activities such as branches, commits, and Pull Requests can be associated with Jira work items.

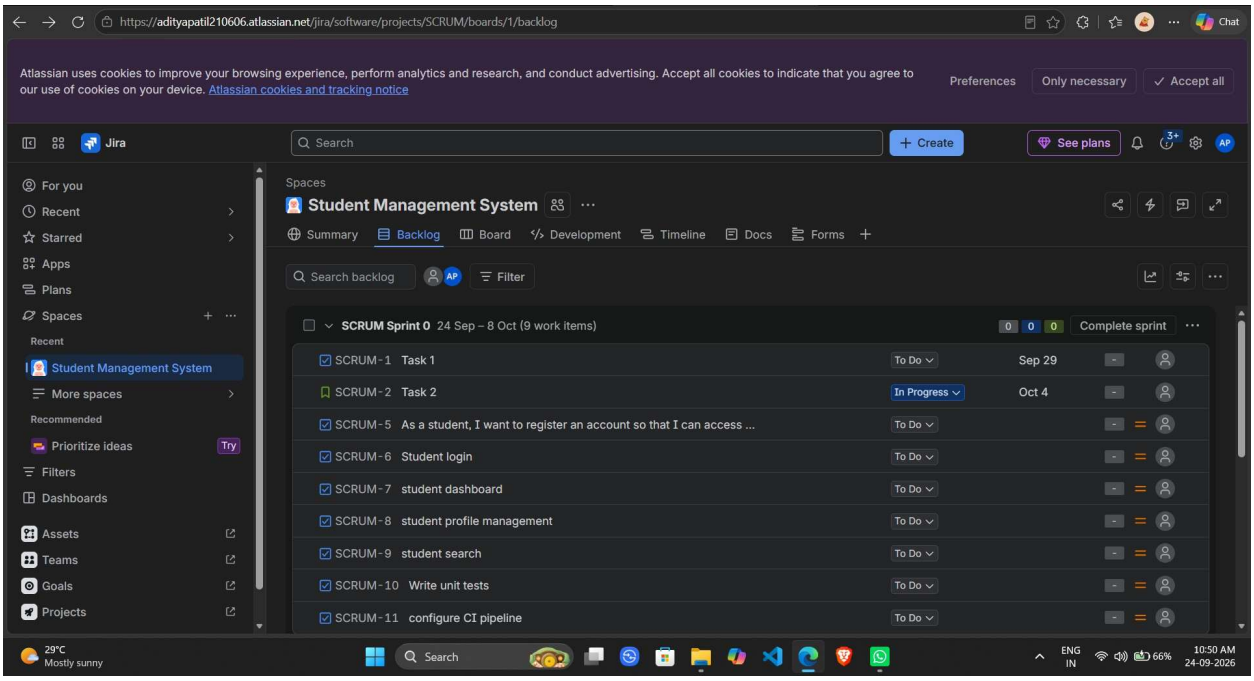
This provides traceability between:

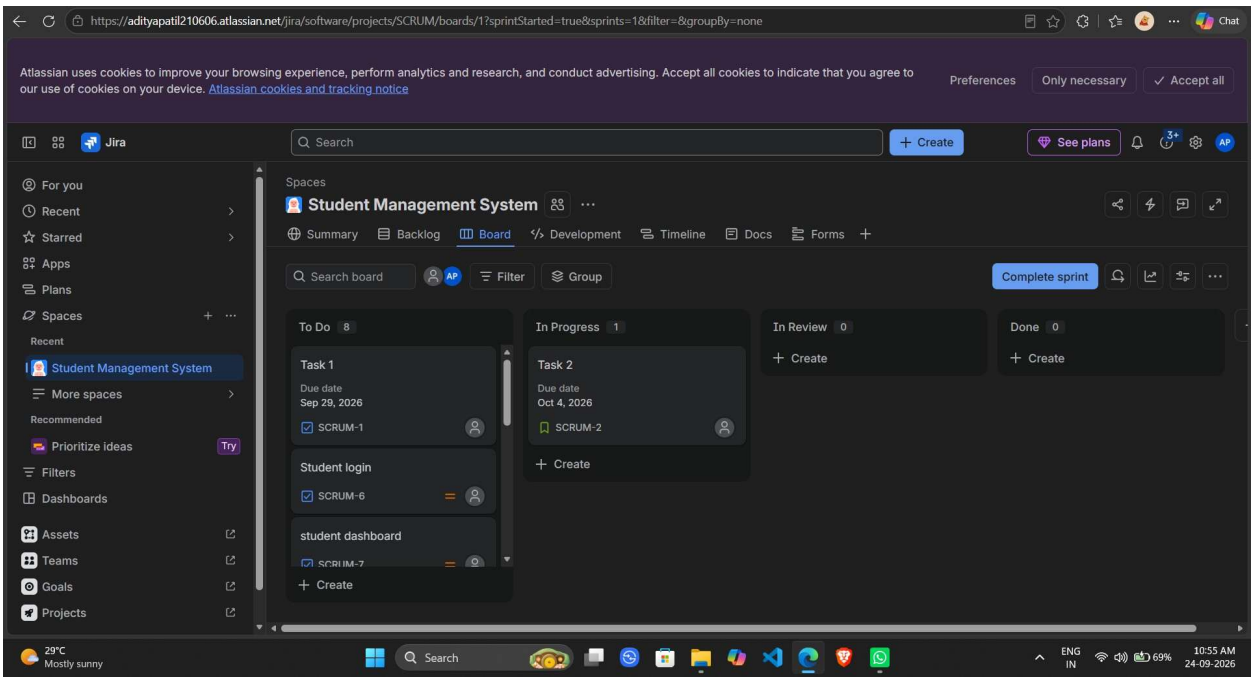
**Jira Work Item → GitHub Branch → Commit → Pull Request → Merge → Jira Status**

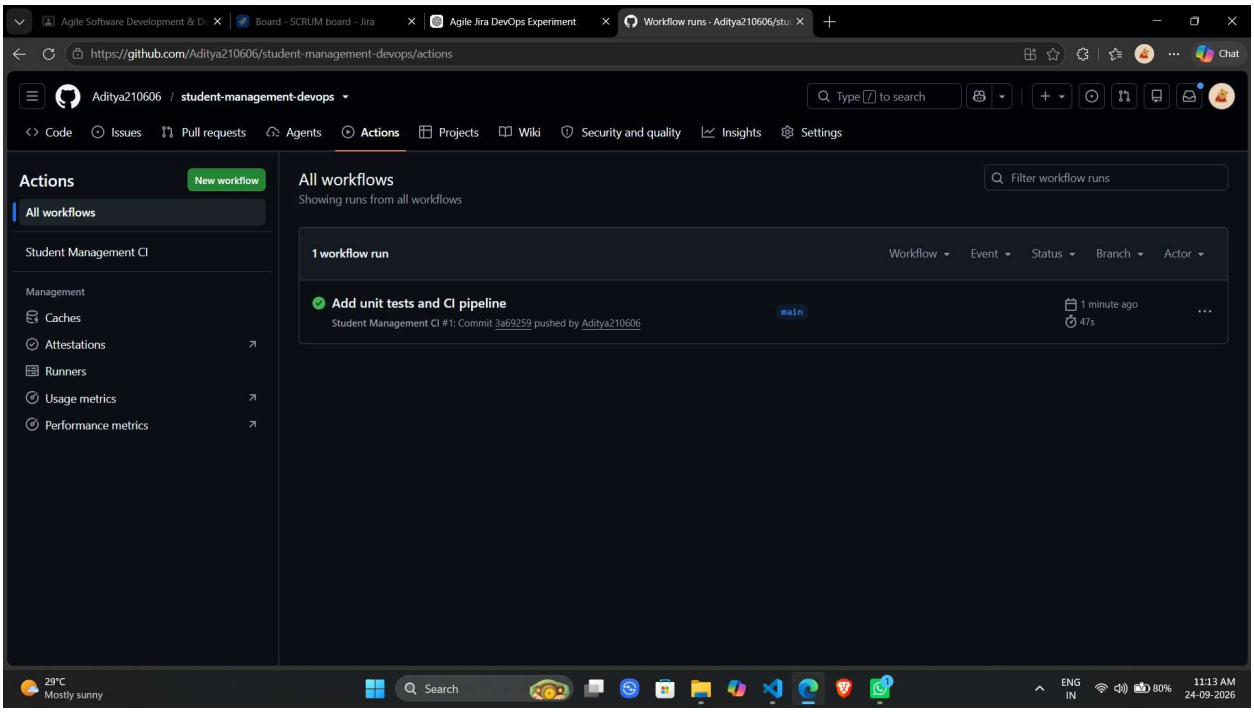
This integration helps the development team monitor both project-management activities and actual code-development activities from Jira.

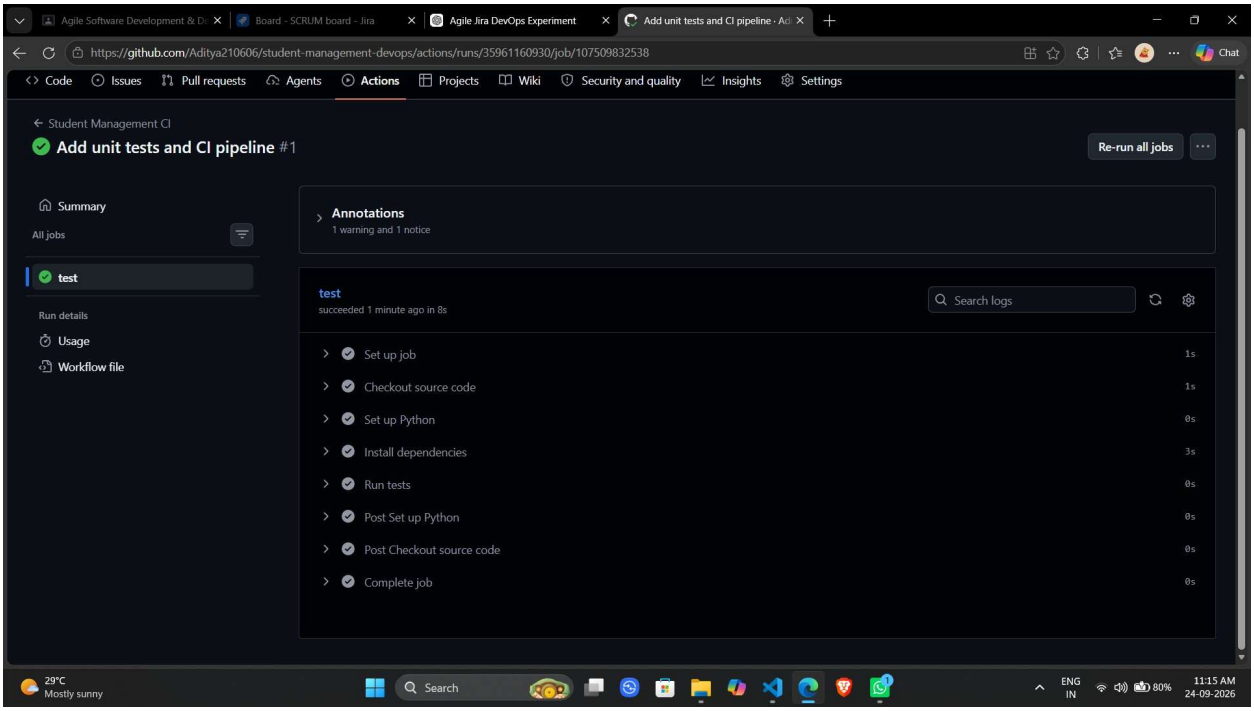
**Output :**

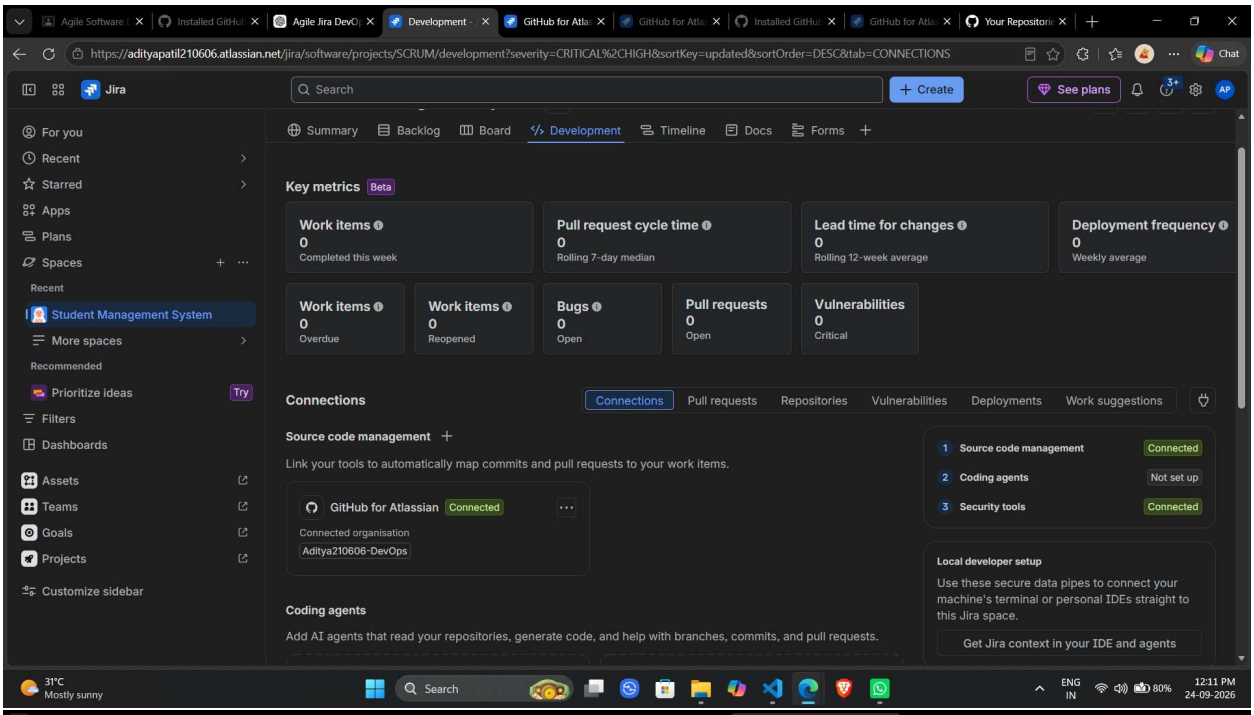


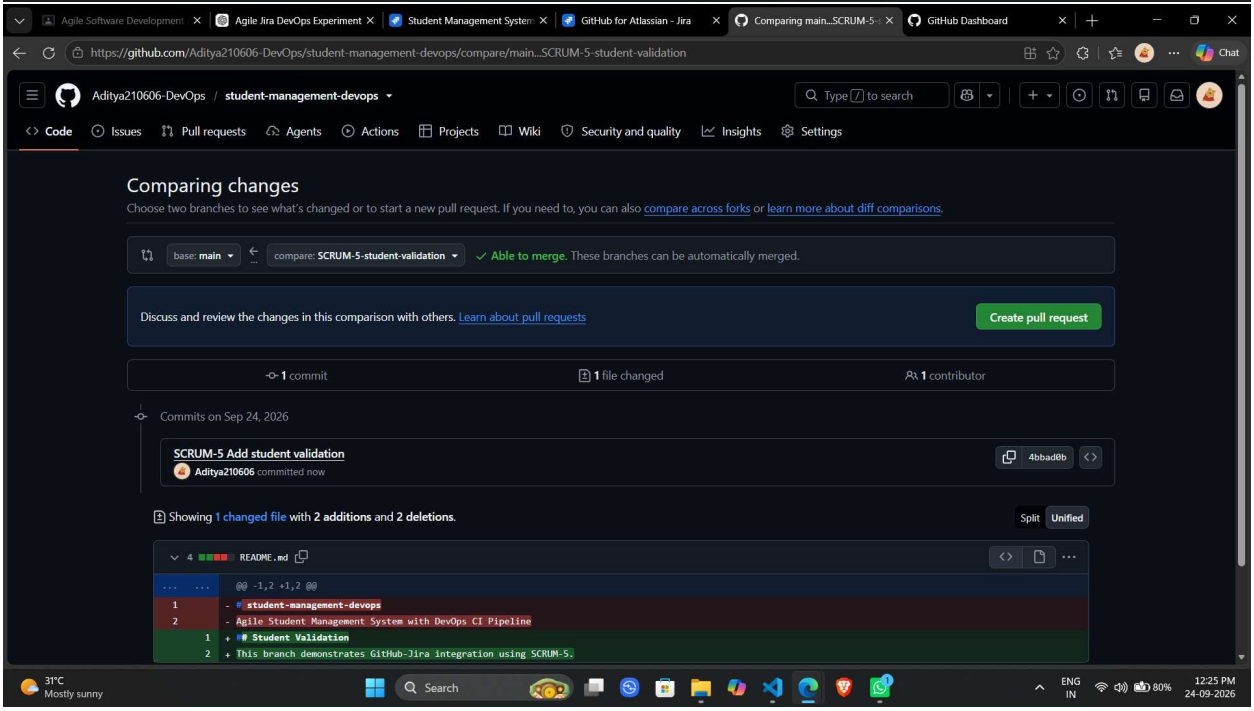

https://adityapatil210606.atlassian.net/jira/software/projects/SCRUM/boards/1/backlog
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Preferences
Only necessary
Accept all
Jira
Search
Create
See plans
For you
Recent
Starred
Apps
Plans
Spaces
Recent
Student Management System
More spaces
Recommended
Prioritize ideas
Try
Filters
Dashboards
Assets
Teams
Goals
Projects
Spaces
Student Management System
Summary
Backlog
Board
Development
Timeline
Docs
Forms
Search backlog
Filter
SCRUM Sprint 0 24 Sep – 8 Oct (9 work items)
Complete sprint
SCRUM-1 Task 1 To Do Sep 29
SCRUM-2 Task 2 In Progress Oct 4
SCRUM-5 As a student, I want to register an account so that I can access ... To Do
SCRUM-6 Student login To Do
SCRUM-7 student dashboard To Do
SCRUM-8 student profile management To Do
SCRUM-9 student search To Do
SCRUM-10 Write unit tests To Do
SCRUM-11 configure CI pipeline To Do
29°C
Mostly sunny
Search
ENG
IN
66%
10:50 AM
24-09-2026

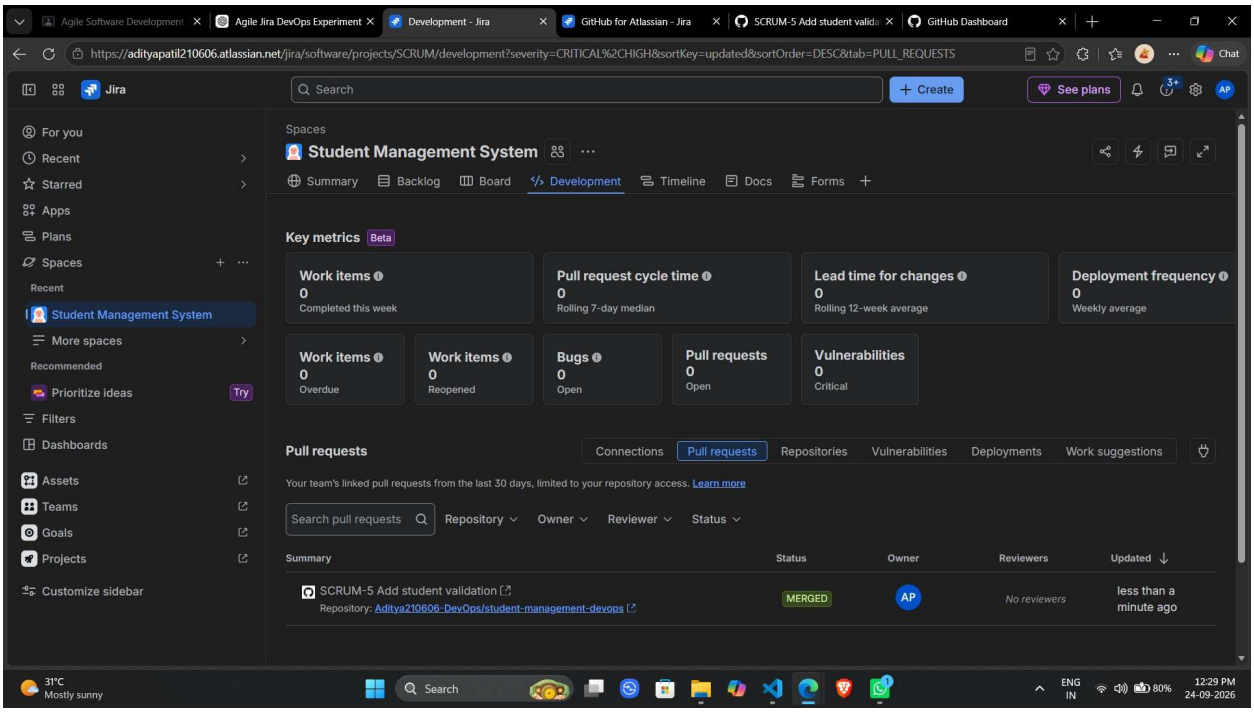

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Preferences
Only necessary
Accept all
Jira
Search
Create
See plans
Spaces
Student Management System
Summary
Backlog
Board
Development
Timeline
Docs
Forms
Search board
Filter
Group
Complete sprint
To Do 8
In Progress 1
In Review 0
Done 0
Task 1
Due date
Sep 29, 2026
SCRUM-1
Task 2
Due date
Oct 4, 2026
SCRUM-2
Student login
SCRUM-6
student dashboard
Create
29°C
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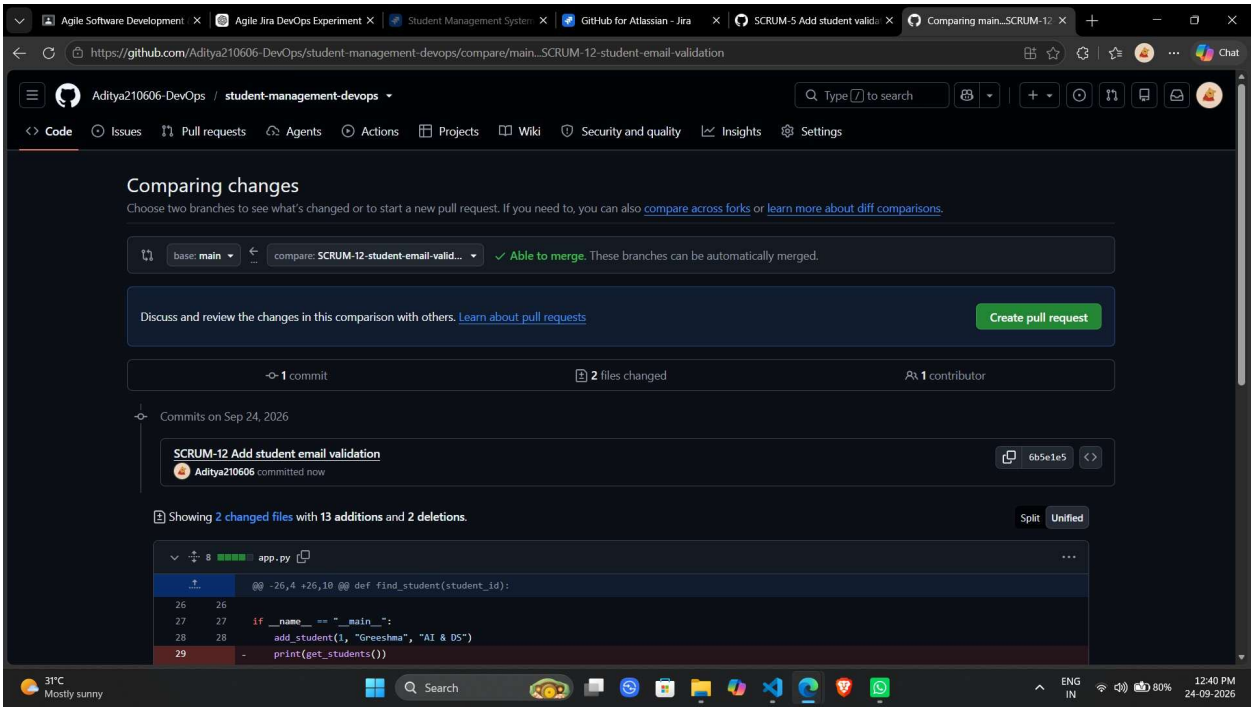

Agile Software Development & Dr
Board - SCRUM board - Jira
Agile Jira DevOps Experiment
Workflow runs · Aditya210606/stu
https://github.com/Aditya210606/student-management-devops/actions
Aditya210606 / student-management-devops
Type / to search
Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
Actions
New workflow
All workflows
Student Management CI
Management
Caches
Attestations
Runners
Usage metrics
Performance metrics
All workflows
Showing runs from all workflows
Filter workflow runs
1 workflow run
Workflow
Event
Status
Branch
Actor
Add unit tests and CI pipeline
Student Management CI #1: Commit 3a69259 pushed by Aditya210606
main
1 minute ago
47s
29°C
Mostly sunny
Search
ENG
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24-09-2026

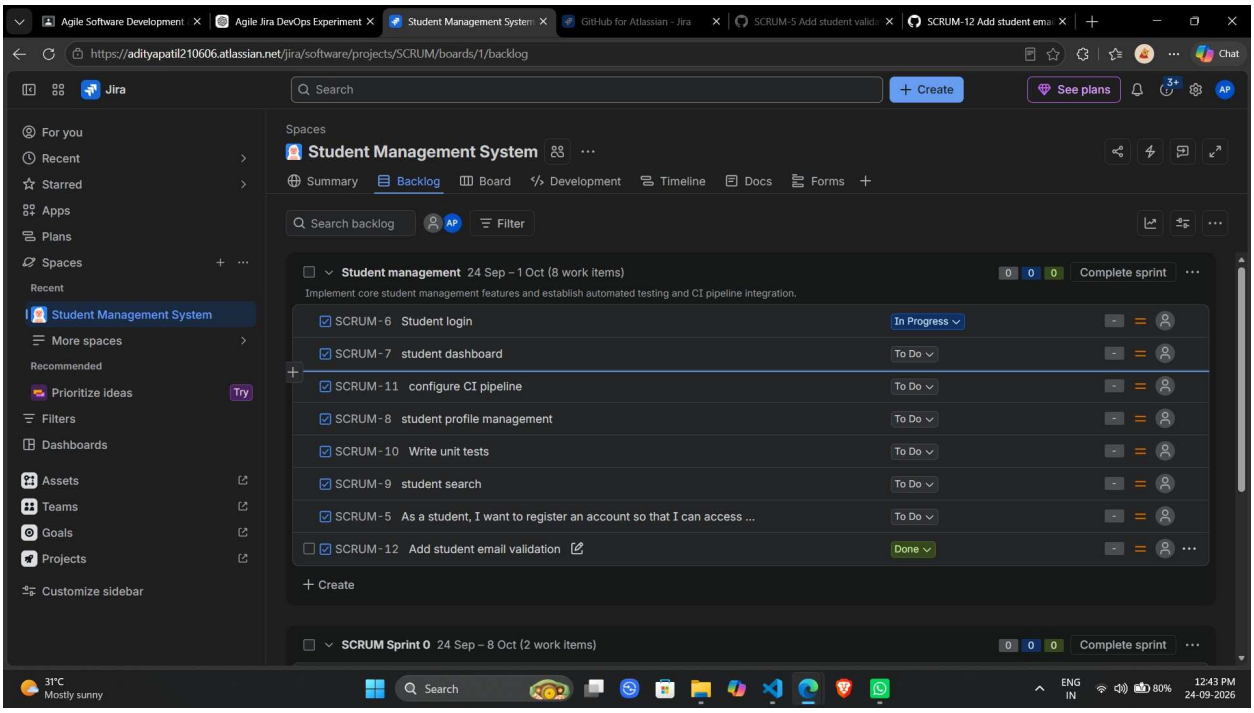

Agile Software Development & Dr
Board - SCRUM board - Jira
Agile Jira DevOps Experiment
Add unit tests and CI pipeline · Ad
https://github.com/Aditya210606/student-management-devops/actions/runs/35961160930/job/107509832538
Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
Student Management CI
Add unit tests and CI pipeline #1
Re-run all jobs
Summary
All jobs
test
Run details
Usage
Workflow file
Annotations
1 warning and 1 notice
test
succeeded 1 minute ago in 8s
Search logs
Set up job
1s
Checkout source code
1s
Set up Python
0s
Install dependencies
3s
Run tests
0s
Post Set up Python
0s
Post Checkout source code
0s
Complete job
0s
29°C
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24-09-2026


https://adityapatil210606.atlassian.net/jira/software/projects/SCRUM/development?severity=CRITICAL%2CHIGH&sortKey=updated&sortOrder=DESC&tab=CONNECTIONS
Jira
Search
Create
See plans
For you
Recent
Starred
Apps
Plans
Spaces
Student Management System
More spaces
Prioritize ideas
Filters
Dashboards
Assets
Teams
Goals
Projects
Customize sidebar
Summary
Backlog
Board
Development
Timeline
Docs
Forms
Key metrics
Beta
Work items
0
Completed this week
Pull request cycle time
0
Rolling 7-day median
Lead time for changes
0
Rolling 12-week average
Deployment frequency
0
Weekly average
Work items
0
Overdue
Work items
0
Reopened
Bugs
0
Open
Pull requests
0
Open
Vulnerabilities
0
Critical
Connections
Connections
Pull requests
Repositories
Vulnerabilities
Deployments
Work suggestions
Source code management
Link your tools to automatically map commits and pull requests to your work items.
GitHub for Atlassian
Connected
Connected organisation
Aditya210606-DevOps
1 Source code management
Connected
2 Coding agents
Not set up
3 Security tools
Connected
Local developer setup
Use these secure data pipes to connect your machine's terminal or personal IDEs straight to this Jira space.
Get Jira context in your IDE and agents
Coding agents
Add AI agents that read your repositories, generate code, and help with branches, commits, and pull requests.
31°C
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12:11 PM
24-09-2026


https://github.com/Aditya210606-DevOps/student-management-devops/compare/main...SCRUM-5-student-validation
Aditya210606-DevOps / student-management-devops
Type / to search
Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
Comparing changes
Choose two branches to see what's changed or to start a new pull request. If you need to, you can also compare across forks or learn more about diff comparisons.
base: main
compare: SCRUM-5-student-validation
Able to merge. These branches can be automatically merged.
Discuss and review the changes in this comparison with others. Learn about pull requests
Create pull request
1 commit
1 file changed
1 contributor
Commits on Sep 24, 2026
SCRUM-5 Add student validation
Aditya210606 committed now
4bbad0b
Showing 1 changed file with 2 additions and 2 deletions.
Split
Unified
README.md
@@ -1,2 +1,2 @@
- # student-management-devops
- Agile Student Management System with DevOps CI Pipeline
+ # Student Validation
+ This branch demonstrates GitHub-Jira integration using SCRUM-5.
31°C
Mostly sunny
Search
ENG
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12:25 PM
24-09-2026


Agile Software Development
Agile Jira DevOps Experiment
Development - Jira
GitHub for Atlassian - Jira
SCRUM-5 Add student valida
GitHub Dashboard
https://adityapatil210606.atlassian.net/jira/software/projects/SCRUM/development?severity=CRITICAL%2CHIGH&sortKey=updated&sortOrder=DESC&tab=PULL_REQUESTS
Jira
Search
Create
See plans
For you
Recent
Starred
Apps
Plans
Spaces
Recent
Student Management System
More spaces
Recommended
Prioritize ideas
Try
Filters
Dashboards
Assets
Teams
Goals
Projects
Customize sidebar
Spaces
Student Management System
Summary
Backlog
Board
Development
Timeline
Docs
Forms
Key metrics
Beta
Work items
0
Completed this week
Pull request cycle time
0
Rolling 7-day median
Lead time for changes
0
Rolling 12-week average
Deployment frequency
0
Weekly average
Work items
0
Overdue
Work items
0
Reopened
Bugs
0
Open
Pull requests
0
Open
Vulnerabilities
0
Critical
Pull requests
Connections
Pull requests
Repositories
Vulnerabilities
Deployments
Work suggestions
Your team's linked pull requests from the last 30 days, limited to your repository access. Learn more
Search pull requests
Repository
Owner
Reviewer
Status
Summary
Status
Owner
Reviewers
Updated
SCRUM-5 Add student validation
Repository: Aditya210606-DevOps/student-management-devops
MERGED
AP
No reviewers
less than a minute ago
31°C
Mostly sunny
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24-09-2026


Agile Software Development
Agile Jira DevOps Experiment
Student Management Syster
GitHub for Atlassian - Jira
SCRUM-5 Add student valida
Comparing main...SCRUM-12
https://github.com/Aditya210606-DevOps/student-management-devops/compare/main...SCRUM-12-student-email-validation
Aditya210606-DevOps / student-management-devops
Type / to search
Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
Comparing changes
Choose two branches to see what's changed or to start a new pull request. If you need to, you can also compare across forks or learn more about diff comparisons.
base: main
compare: SCRUM-12-student-email-valid...
Able to merge. These branches can be automatically merged.
Discuss and review the changes in this comparison with others. Learn about pull requests
Create pull request
1 commit
2 files changed
1 contributor
Commits on Sep 24, 2026
SCRUM-12 Add student email validation
Aditya210606 committed now
6b5e1e5
Showing 2 changed files with 13 additions and 2 deletions.
Split
Unified
app.py
@@ -26,4 +26,10 @@ def find_student(student_id):
if __name__ == "__main__":
add_student(1, "Greeshma", "AI & DS")
print(get_students())
31°C
Mostly sunny
Search
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80%
12:40 PM
24-09-2026




## Conclusion

The Agile Software Project Lifecycle was successfully studied and implemented using Jira with DevOps pipeline integration. Scrum work items were created and tracked in Jira, while GitHub was used for version control, feature branches, commits, and Pull Requests. The completed changes were merged into the main branch, and the development activities were reflected in Jira.

A Continuous Integration pipeline was also configured using GitHub Actions to automate the verification and testing of the Python application. Thus, the experiment demonstrated the complete Agile and DevOps workflow from task management and development to CI execution, Pull Request, code merging, and Jira tracking.

## Questions to be answered by students

### 1. What is the role of Jira in Agile software project management?

**Answer:** Jira is used to plan, track, and manage Agile software projects. It helps teams create user stories and tasks, organize them into sprints, monitor progress using Scrum boards, and manage issues throughout the software development lifecycle.

### 2. What is the purpose of integrating a DevOps CI pipeline with Jira?

**Answer:** Integrating a CI pipeline with Jira connects project management activities with software development and testing. Code changes can be linked to Jira work items, while automated builds and tests verify the changes. This improves traceability, automation, collaboration, and continuous integration.